

7. [Maximum mark: 8]

The discrete random variable X has the following probability distribution:

x	1	2	3
$P(X = x)$	$0.6 - 2a$	$3a$	$0.4 - a$

(a) Find the range of possible values of a . [2]

(b) In the case where $a = 0.2$, determine $\text{Var}(2 - X)$. [6]

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